

## module3\_lab

September 20, 2026

```
[1]: import pandas as pd
df =pd.read_csv("Downloads/messy_sales.csv")
```

```
[2]: df.head(10)
```

```
[2]:
```

|   | order_id | date       | product  | price | qty | zip   |
|---|----------|------------|----------|-------|-----|-------|
| 0 | 1254     | 04/06/2026 | mug      | 11.36 | 4   | 60614 |
| 1 | 1057     | 2026-05-30 | charger  | 14.79 | 3   | 30303 |
| 2 | 1150     | 05/06/2026 | mug      | 5.50  | 1   | 10001 |
| 3 | 1066     | 05/30/2026 | notebook | NaN   | 1   | 98101 |
| 4 | 1114     | 04/06/2026 | webcam   | 45.59 | 4   | 90405 |
| 5 | 1280     | 04/22/2026 | pen set  | NaN   | 4   | 30303 |
| 6 | 1204     | 04/18/2026 | charger  | 8.41  | 2   | 2134  |
| 7 | 1249     | 2026-05-15 | keyboard | 75.04 | 4   | 90405 |
| 8 | 1037     | 2026-05-19 | mug      | 9.31  | 2   | 60614 |
| 9 | 1177     | 2026-05-13 | mug      | 10.11 | 3   | 90210 |

```
[3]: df.head()
```

```
[3]:
```

|   | order_id | date       | product  | price | qty | zip   |
|---|----------|------------|----------|-------|-----|-------|
| 0 | 1254     | 04/06/2026 | mug      | 11.36 | 4   | 60614 |
| 1 | 1057     | 2026-05-30 | charger  | 14.79 | 3   | 30303 |
| 2 | 1150     | 05/06/2026 | mug      | 5.50  | 1   | 10001 |
| 3 | 1066     | 05/30/2026 | notebook | NaN   | 1   | 98101 |
| 4 | 1114     | 04/06/2026 | webcam   | 45.59 | 4   | 90405 |

```
[4]: df.shape
```

```
[4]: (300, 6)
```

```
[5]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 300 entries, 0 to 299
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   order_id    300 non-null    int64
1   date        300 non-null    str
```

```

2  product    300 non-null    str
3  price      288 non-null    float64
4  qty        300 non-null    int64
5  zip        300 non-null    int64
dtypes: float64(1), int64(3), str(2)
memory usage: 19.2 KB

```

```
[6]: #Step 1 for shape, zip to be seen as int64, and date is str
```

```
[7]: df.describe()
```

```

[7]:
   count  order_id  price  qty  zip
count    300.000000  288.000000  300.000000  300.000000
mean    1144.816667   56.068438   2.913333  45385.276667
std       84.515319   72.307182   1.560107  39345.608641
min     1000.000000    2.930000  -5.000000   2116.000000
25%     1071.750000   10.945000   2.000000   2134.000000
50%     1144.500000   37.530000   3.000000  30303.000000
75%     1217.250000   61.417500   4.000000  90210.000000
max     1291.000000  479.290000   5.000000  98101.000000

```

```
[8]: import seaborn as sns
```

```
[9]: df.isnull().sum()
```

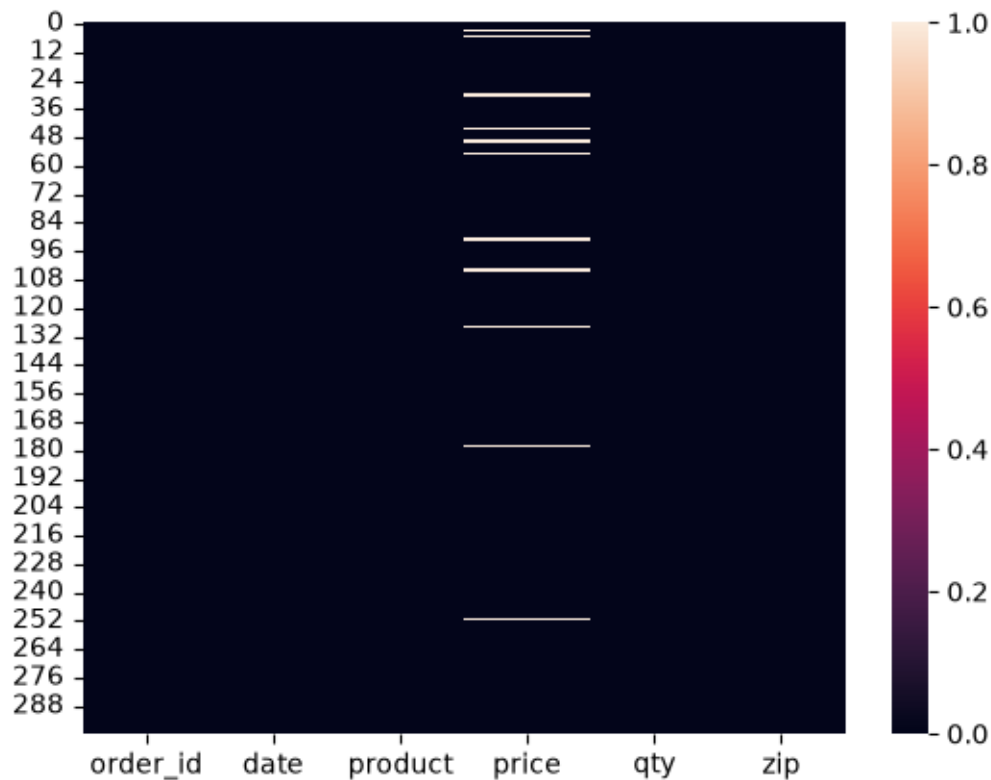
```

[9]: order_id    0
     date       0
     product    0
     price     12
     qty       0
     zip       0
     dtype: int64

```

```
[10]: sns.heatmap(df.isnull())
```

```
[10]: <Axes: >
```



```
[11]: #Running the analysis can see that there is missing values in the price,
      ↳category, 12 to be exact, Also the zip and date are in the incoorect format,
      ↳and running describe saw that there is a negative quantity value of -5 as a
      ↳minimum which are impossible.
```

```
[12]: df.isna().sum()
```

```
[12]: order_id    0
      date        0
      product     0
      price      12
      qty         0
      zip         0
      dtype: int64
```

```
[13]: #step 3 count the missing values
```

```
[14]: fill_value=df["price"].median()
      df["price"]=df["price"].fillna(fill_value)
```

```
[15]: df.isna().sum()
```

```
[15]: order_id    0
      date       0
      product    0
      price      0
      qty        0
      zip        0
      dtype: int64
```

```
[16]: #checkpoint for df.isna.sum()
```

```
[17]: replaced_rows=df[df["price"]==fill_value]
      print ("Rows contain the fill value:")
      print(replaced_rows)
```

Rows contain the fill value:

|     | order_id | date       | product    | price | qty | zip   |
|-----|----------|------------|------------|-------|-----|-------|
| 3   | 1066     | 05/30/2026 | notebook   | 37.53 | 1   | 98101 |
| 5   | 1280     | 04/22/2026 | pen set    | 37.53 | 4   | 30303 |
| 30  | 1088     | 04/28/2026 | pen set    | 37.53 | 2   | 2134  |
| 44  | 1101     | 2026-05-12 | keyboard   | 37.53 | 3   | 2116  |
| 49  | 1199     | 2026-04-18 | charger    | 37.53 | 5   | 30303 |
| 50  | 1018     | 05/04/2026 | monitor    | 37.53 | 4   | 10001 |
| 55  | 1170     | 04/04/2026 | monitor    | 37.53 | 1   | 2116  |
| 91  | 1040     | 05/08/2026 | pen set    | 37.53 | 3   | 2134  |
| 104 | 1240     | 04/16/2026 | headphones | 37.53 | 1   | 60614 |
| 128 | 1261     | 2026-05-09 | desk lamp  | 37.53 | 2   | 90405 |
| 178 | 1005     | 2026-05-24 | monitor    | 37.53 | 2   | 2134  |
| 251 | 1155     | 2026-04-08 | charger    | 37.53 | 3   | 2134  |

```
[18]: #step 4 print filled median values
```

```
[19]: df.duplicated().sum()
```

```
[19]: np.int64(8)
```

```
[20]: #count duplicates where there are 8
```

```
[21]: df.shape
```

```
[21]: (300, 6)
```

```
[22]: df.drop_duplicates()
```

```
[22]:
```

|   | order_id | date       | product  | price | qty | zip   |
|---|----------|------------|----------|-------|-----|-------|
| 0 | 1254     | 04/06/2026 | mug      | 11.36 | 4   | 60614 |
| 1 | 1057     | 2026-05-30 | charger  | 14.79 | 3   | 30303 |
| 2 | 1150     | 05/06/2026 | mug      | 5.50  | 1   | 10001 |
| 3 | 1066     | 05/30/2026 | notebook | 37.53 | 1   | 98101 |

```

4          1114  04/06/2026    webcam  45.59    4  90405
..          ...          ...          ...    ...    ..
294        1103  2026-04-21    pen set  16.47    3  90405
296        1067  2026-04-27    backpack  37.93    5  90210
297        1025  2026-04-27    keyboard  47.30   -2   2116
298        1196  04/25/2026    pen set   8.79    3  90405
299        1175  2026-05-16         mug   4.63    4  10001

```

```
[292 rows x 6 columns]
```

```
[23]: #removed duplicates and values show 292x6
```

```
df["zip"]=df["zip"].astype(str).str.zfill(5)df.info()
```

```
[24]: df.info()
```

```

<class 'pandas.DataFrame'>
RangeIndex: 300 entries, 0 to 299
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   order_id    300 non-null    int64
1   date        300 non-null    str
2   product     300 non-null    str
3   price       300 non-null    float64
4   qty         300 non-null    int64
5   zip         300 non-null    int64
dtypes: float64(1), int64(3), str(2)
memory usage: 19.2 KB

```

```
[25]: df["zip"]=df["zip"].astype(str).str.zfill(5)
```

```
[26]: #step 6 repair the zip code type and verify
```

```
[27]: df.info()
```

```

<class 'pandas.DataFrame'>
RangeIndex: 300 entries, 0 to 299
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   order_id    300 non-null    int64
1   date        300 non-null    str
2   product     300 non-null    str
3   price       300 non-null    float64
4   qty         300 non-null    int64
5   zip         300 non-null    str
dtypes: float64(1), int64(2), str(3)
memory usage: 20.7 KB

```

```
[28]: df.shape
```

```
[28]: (300, 6)
```

```
[29]: #found chart was not 292,6 had to drop duplicates instead of running it in  
↳steps earlier
```

```
[30]: df = df.drop_duplicates()
```

```
[31]: df.drop_duplicates()
```

```
[31]:
```

|     | order_id | date       | product  | price | qty | zip   |
|-----|----------|------------|----------|-------|-----|-------|
| 0   | 1254     | 04/06/2026 | mug      | 11.36 | 4   | 60614 |
| 1   | 1057     | 2026-05-30 | charger  | 14.79 | 3   | 30303 |
| 2   | 1150     | 05/06/2026 | mug      | 5.50  | 1   | 10001 |
| 3   | 1066     | 05/30/2026 | notebook | 37.53 | 1   | 98101 |
| 4   | 1114     | 04/06/2026 | webcam   | 45.59 | 4   | 90405 |
| ..  | ...      | ...        | ...      | ...   | ... | ...   |
| 294 | 1103     | 2026-04-21 | pen set  | 16.47 | 3   | 90405 |
| 296 | 1067     | 2026-04-27 | backpack | 37.93 | 5   | 90210 |
| 297 | 1025     | 2026-04-27 | keyboard | 47.30 | -2  | 02116 |
| 298 | 1196     | 04/25/2026 | pen set  | 8.79  | 3   | 90405 |
| 299 | 1175     | 2026-05-16 | mug      | 4.63  | 4   | 10001 |

```
[292 rows x 6 columns]
```

```
[32]: df.shape
```

```
[32]: (292, 6)
```

```
[33]: #confirmed new chart has correct shape
```

```
[34]: df.info()
```

```
<class 'pandas.DataFrame'>  
Index: 292 entries, 0 to 299  
Data columns (total 6 columns):  
#   Column      Non-Null Count  Dtype  
---  -  
0   order_id    292 non-null    int64  
1   date        292 non-null    str  
2   product     292 non-null    str  
3   price       292 non-null    float64  
4   qty         292 non-null    int64  
5   zip         292 non-null    str  
dtypes: float64(1), int64(2), str(3)  
memory usage: 22.4 KB
```

```
[35]: df["date"]=pd.to_datetime(df["date"],format="mixed")
```

```
[36]: #standarized dates and checked below with info
```

```
[37]: df.info()
```

```
<class 'pandas.DataFrame'>
Index: 292 entries, 0 to 299
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   order_id    292 non-null    int64
1   date        292 non-null    datetime64[us]
2   product     292 non-null    str
3   price       292 non-null    float64
4   qty         292 non-null    int64
5   zip         292 non-null    str
dtypes: datetime64[us](1), float64(1), int64(2), str(2)
memory usage: 19.5 KB
```

```
[38]: df.shape
```

```
[38]: (292, 6)
```

```
[39]: df.head(10)
```

```
[39]:
```

|   | order_id | date       | product  | price | qty | zip   |
|---|----------|------------|----------|-------|-----|-------|
| 0 | 1254     | 2026-04-06 | mug      | 11.36 | 4   | 60614 |
| 1 | 1057     | 2026-05-30 | charger  | 14.79 | 3   | 30303 |
| 2 | 1150     | 2026-05-06 | mug      | 5.50  | 1   | 10001 |
| 3 | 1066     | 2026-05-30 | notebook | 37.53 | 1   | 98101 |
| 4 | 1114     | 2026-04-06 | webcam   | 45.59 | 4   | 90405 |
| 5 | 1280     | 2026-04-22 | pen set  | 37.53 | 4   | 30303 |
| 6 | 1204     | 2026-04-18 | charger  | 8.41  | 2   | 02134 |
| 7 | 1249     | 2026-05-15 | keyboard | 75.04 | 4   | 90405 |
| 8 | 1037     | 2026-05-19 | mug      | 9.31  | 2   | 60614 |
| 9 | 1177     | 2026-05-13 | mug      | 10.11 | 3   | 90210 |

```
[40]: df[df["qty"]<0]
```

```
[40]:
```

|     | order_id | date       | product   | price | qty | zip   |
|-----|----------|------------|-----------|-------|-----|-------|
| 202 | 1140     | 2026-04-18 | webcam    | 38.69 | -5  | 98101 |
| 262 | 1233     | 2026-05-09 | desk lamp | 22.64 | -4  | 10001 |
| 297 | 1025     | 2026-04-27 | keyboard  | 47.30 | -2  | 02116 |

```
[41]: df.loc[df["qty"]<0,"qty"] = df["qty"].abs()
```

```
[42]: df.loc[[202,262,297]]
```

```
[42]:
```

|  | order_id | date | product    | price     | qty   | zip     |
|--|----------|------|------------|-----------|-------|---------|
|  | 202      | 1140 | 2026-04-18 | webcam    | 38.69 | 5 98101 |
|  | 262      | 1233 | 2026-05-09 | desk lamp | 22.64 | 4 10001 |
|  | 297      | 1025 | 2026-04-27 | keyboard  | 47.30 | 2 02116 |

```
[43]: df.to_csv("sales_clean.csv",index=False)
```

```
[44]: #Cleaning Log
#300 loaded; 12 prices filled with 37.53; 8 duplicates removed leaving 292;␣
  ↳ zips restored; dates standardized; negatives handled by having them become␣
  ↳ absolute values as they can't be negative qty
# This can lead to wrong choice since duplicate values can skew the data and␣
  ↳ create bias towards making a decision.
```